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Summary

I am an experienced Full-Stack Developer, Mentor/Tutor, UI/UX Designer, and Retail/Sales Manager driven by curiosity and a passion for continuous learning. While focused on personal growth, I am pursuing a PhD in Computer Science with a focus on Human-Computer Interaction (HCI), intending to combine it with medicine.

Professional Experience & Research

Colorado Sustainability Hub, Graduate Research Assistant

09.2025 – Present

University of Denver | Website: SustainabilityHub.co

Denver, CO

- Contributing to the NSF-funded Sustainability Hub, an interdisciplinary project advancing sustainable data access, visualization, and inclusive AI interaction design.
- Designing and refining UI/UX systems, visual assets, and chatbot interaction models in Figma to enhance public engagement and data understanding.
- Supporting the research-driven design and prototyping of interactive web interfaces for sustainability communication, with emphasis on accessibility, inclusivity, and human-AI collaboration.
- Co-authoring the forthcoming paper “Evaluating LLMs in Context: A Survey of Metrics, Gaps, and Domain-Specific Benchmarks”, which supports the Hub’s evaluation framework for sustainability-focused AI systems.
- Conducting independent research in accessible and inclusive design of large language model interfaces, aimed at improving usability across technical and non-technical users.
- Assisting with website iteration, PR reviews, and design consistency across interdisciplinary teams.

Roadrunner Connect, Full Stack Mobile App Development Project Lead & Intern

06.2024 – 07.2025

Metropolitan State University of Denver | Website: RoadrunnerConnect.co

Denver, CO

- Led and developed 'Roadrunner Connect,' an interactive campus engagement app.
- Built frontend with Flutter (Dart) for mobile and React (JavaScript/CSS) for the web; developed backend using MongoDB/Node.js hosted on AWS for scalability and reliability. Utilized NLP (Natural Language Processing) to tag events appropriately based on their descriptions using predefined tags.
- We integrated Google Maps API and Google Geolocation API for the dynamic, interactive map view and location-based features. The system leverages AWS services to host our MongoDB database and run our event parsing scripts, ensuring scalability and reliability.
- Integrated Firebase for authentication (Google/iOS) and Gravatar for the team's profiles.
- Implemented features such as event check-ins, user-uploaded content (photos/comments), badges with an earned animation, challenges, a feather confetti animation, a rewards system (with giveaways and leaderboards), varied viewing modes (list/map/calendar), and a social friends system.
- Collaborated in an Agile team as the Scrum Master, gaining leadership skills, managing task distribution, progress tracking, and Agile workflows using ZenHub; oversaw team Objectives and Key Results (OKRs) to align development with strategic goals; handled version control using Git/GitHub.
- Regularly presented app progress and independently represented the team in stakeholder meetings (15+ attendees), streamlined communications, and incorporated stakeholder feedback into iterative improvements.

- Directed marketing strategies and promotional materials, leading outreach to maximize campus-wide adoption.
- Prioritized user accessibility, perspective, usability (UI/UX), and human-centered design throughout the development process in Figma.
- Authored the abstract for the 2025 Undergraduate Research Conference (URC) at MSU Denver, showcasing the impact of the app and its evolution.
- Facilitated user testing sessions, beta launch events, and stress-test events, ensuring comprehensive user feedback and continuous improvement.
- Attended and presented at numerous promotional and community events, including CS Community Hours, sports games, campus outreach events, user engagement activities, and conferences.

Metropolitan State University of Denver, Learning Assistant (LA) — Computer Science 1

08.2024 – 12.2024
Denver, CO

- I was endorsed by multiple professors to serve as an LA for CS1, providing comprehensive support to 30+ students in fundamental programming concepts and problem-solving techniques.
- Conducted weekly one-on-one tutoring sessions and held 10 office hours weekly, contributing to a 20% decrease in late project submissions.
- Assisted in lectures and provided supplementary learning materials, offering hands-on support for students implementing concepts in practical scenarios.
- Acted as a tutor for other CS courses, leveraging broad knowledge to support students across various areas of computer science.

Colorado Sustainability Hub, Undergraduate Research Assistant
Metropolitan State University of Denver | Website: [SustainabilityHub.co](https://sustainabilityhub.co)

09.2023 – 06.2024
Denver, CO

- Contributed to the Sustainability Hub, a National Science Foundation (NSF) funded initiative focused on sustainability data analysis, application prototyping, and machine learning model development.
- Collaborated with a team of 6 researchers in an Agile environment to create UI/UX designs and implement a prototype for the Sustainability Hub application.
- Developed machine learning models to identify patterns within sustainability data, enhancing the project's objectives towards sustainability, data democracy, and inclusivity.
- Worked under the guidance of the project manager to advance the initiative's goals in sustainability data analysis and application development.
- Co-authored the abstract and presented the project and its findings at the 2024 Undergraduate Research Symposium for MSU Denver, showcasing the team's progress and impact.

GameStop Inc., Assistant Store Leader (ASL)

04.2023 – 06.2024
Aurora/Denver, CO

- Assisted in all aspects of store operations, including inventory management, payroll, scheduling, and maintaining the store's Profit and Loss (P&L), while consistently leading the district in sales performance and exceeding KPI goals.
- Led and motivated a team to provide exceptional customer service, implemented effective merchandising strategies (including planogram execution and visual displays to drive sales), and managed opening/closing procedures, financial record-keeping, and loss-prevention measures.
- Collaborated with district management on multi-unit tasks, including inventory preparation, staff coaching at other locations, and contributing to the creation of a company-wide adopted business sale method.
- Demonstrated expertise in loss prevention, reducing shrink through analysis, and product placement optimization, while adhering to local pawn laws for product trades.
- Excelled in individual performance metrics by leading the district in our sales metrics for most quarters in 2024, maintained flexibility in covering shifts and coordinating district-wide coverage, and assisted in talent sourcing, onboarding, and development across the district.

Metropolitan State University of Denver , <i>Learning Assistant (LA) - Discrete Structures</i>	01.2024 – 05.2024 Denver, CO
• Provided comprehensive tutoring and academic support to 30+ students in Discrete Structures, leveraging a personal passion for the subject to make complex concepts accessible and engaging. • Conducted weekly one-on-one tutoring sessions and held 10 office hours weekly, helping increase student exam scores by an average of 12%. • Gained a deeper understanding of the material by having assisted a different professor than the one I had taken the course with, broadening my knowledge of various teaching approaches and content coverage. • Crafted innovative teaching strategies to help students overcome common obstacles in understanding abstract mathematical concepts, demonstrating real-world applications in the field. • Finished the semester with a team of LAs to create and present a project on how the LA experience was, and what it taught us.	
Elitch Gardens Amusement Park, Concession Staffing Stands (CSS) , <i>Supervisor</i>	07.2021 – 05.2024 Denver, CO
• Coordinating myself and fellow employees throughout each of our (four) stands. • Operating a cash register, grill, and fryer. Therefore mastering POS systems to handle payments. • Taking charge in conflict resolution, resulting in customer satisfaction.	

Education

Doctor of Philosophy in Computer Science , <i>University of Denver (DU)</i>	09.2025 – 05.2031 Denver, CO
Bachelor of Science in Computer Science , <i>Metropolitan State University of Denver (MSU)</i>	01.2022 – 12.2024 Denver, CO
• GPA: 3.74 Cum Laude • Honors: • Graduated Cum Laude • Provost's Honor Roll • President's Honor Roll	
Bachelor of Science in Computer Science , <i>University of Colorado Boulder (CU Boulder)</i>	08.2021 – 12.2021 Denver, CO
East High School	08.2017 – 05.2021 Denver, CO
• GPA: 3.8 • Honor: • Principal's Honor Roll • Denver East Academic Letter – The highest standard of academic excellence	

Awards, Honors & Certifications

DU/MSU/CU Denver Hackathon 1st Place Award	April, 2025
• Competed in the team 'Bit Happens,' developing 'Lucent,' an AI-driven mental wellness mobile app designed to enhance campus life through immediate emotional support and tailored resource connections. • Built continuously over 26+ hours, staying awake and focused throughout the entire event. • Engineered frontend features using Flutter, implemented backend APIs with FastAPI, and managed data storage using Supabase (PostgreSQL). • Integrated GPT-4.1 for a wellness chatbot, enabling real-time reflective interactions based on peer-reviewed therapeutic methods, validated by Rutgers University research. • Implemented an engaging visual streak system using an interactive bonsai tree, fostering daily emotional reflection and user retention.	

DU/MSU Hackathon 2nd Place Award

October, 2023

- Participated with my team, 'Bits Please', to create 'SpiderByte,' a coding challenge website for classroom learning (similar to Leetcode, but more designed to align with a course's material).
- This was created during a 28+ hour hackathon, after staying awake for 36+ hours, and overall winning 2nd place among 6+ other teams.
- Tools & Technologies: Python, Flask, SQLAlchemy, HTML/CSS, PythonTutor, and CodeMirror.
- Enhanced user experience with PythonTutor for code visualization and CodeMirror for in-browser editing.
- Established and set up test cases to validate user-submitted code, ensuring accuracy and precision.

Certificate: Fundamentals of Deep Learning with NVIDIA

December, 2023

- Completing NVIDIA's Deep Learning course, helped me gain expertise in neural network fundamentals and practical implementation.
- I also learned how to use TensorFlow and Keras to build deep learning models.

Certificate: SQL Competition Winner

April, 2024

- I won first place in MSU Denver's SQL competition for Principles of Database Systems (CS 3810) against 30+ other students.
- I became the first woman to win this competition, making history in a generally male-dominated field.

Honors

- Cum Laude, UPE Honors Society, Provost's Honor Roll, President's Honor Roll, Principal's Honor Roll, and Denver East Academic Letter.

UPE (Upsilon Pi Epsilon Honors Society) Outstanding Achievement Scholarship Award

December, 2024

DSF (Denver Scholarship Foundation) Scholarship

September, 2021

- Recipient of the Denver Scholarship Foundation (DSF) Scholarship, awarded for academic achievement and financial need. This scholarship is regularly reevaluated to ensure I meet the program's ongoing academic and financial requirements, providing continuous support throughout my undergraduate education.

East Angels Friends & Alumni Foundation One-Time Scholarship

May, 2021

Notable Projects

Independent Study – Deep Learning Educational Tool

08.2024 – 12.2024

- Developed an educational chatbot in a team of three using RAG (Retrieval-Augmented Generation) to help students and professors engage with quiz material from Discrete Structures.
- Independently, each team member began exploring the basics, later combining ideas to create a cohesive tool.
- The chatbot uses textbook data to help students review quizzes, understand mistakes, and receive tailored follow-up questions for extra credit based on quiz structure.
- **Core Features:**
 - Integrated Ollama (Llama 3) as a local large language model for chatbot interactions.
 - Processed textbook PDFs using OpenAI embeddings, SKLearnVectorStores, and PyPDF2 for accurate retrieval of examples, diagrams, and context.
 - Built with Streamlit for a user-friendly interface and focused on improving educational impact by designing the chatbot to guide users to solutions rather than providing direct answers.
- **Tools & Technologies:**

- Python, Streamlit, LangChain, RAG pipeline, OpenAI embeddings, and PyPDF2.
- Data Management: Chunking and tokenization for efficient vector searches and semantic matching.

• **Project Goals / Next Steps:**

- Expand course coverage to include other courses like Algorithms.
- Conduct user testing for a potential research publication on the tool's impact on education.
- Integration with the Canvas API to allow login and class/quiz retrieval.

Senior Experience Project – SpiderByte, Hackathon Continuation

08.2024 – 12.2024

- Built upon the prototype developed during a Hackathon in October 2023, where the project won 2nd Place, restarting development from scratch to align with user-centered design principles.
- Collaborated in a 5-person team over 15+ weeks, using stakeholder feedback and iterative presentations (4+) to refine features, improve user experience, and align the product with user needs.
- Redeveloped *SpiderByte* as a full-stack web application:
 - **Frontend:** React with an emphasis on UI/UX and accessibility.
 - **Backend:** ExpressJS, MongoDB, and Mongoose for API development and data management.
 - **Key Features:** Monaco Editor for in-browser Python coding; Docker and RabbitMQ for sandboxing and asynchronous processing; Node WebSocket Server for session management; PythonTutor for syntax tree visualization; Gemini and a Sentence Transformers NLP (Natural Language Processing) model for generating and similarity scoring AI-generated coding challenges.
- Emphasized learning and exploration, while considering future features, including a progress visualization page ("Roadmap"), additional language support, and interactive spider animations to enhance engagement.
- **Tools & Technologies:** React, Figma, Git/GitHub, Zenhub, Docker, Python, SharePoint, Postman, PythonTutor, MongoDB, Monaco Editor, Gemini, RabbitMQ, ExpressJS, Sentence Transformers.

Conferences & Presentations

2025 Undergraduate Research Conference (URC),

April, 2025

Metropolitan State University of Denver

- Presented as the project lead. Authored a research abstract on Roadrunner Connect and presented a poster highlighting its development, impact, evolution, and future directions.

Faculty Research Symposium (FRS), Metropolitan State University of Denver

October, 2024

- Co-presented Roadrunner Connect, revealing the app's development progress and expressing appreciation to our stakeholders. Aimed to raise awareness and strengthen support.

2024 Undergraduate Research Conference (URC),

April, 2024

Metropolitan State University of Denver

- Demonstrated the Sustainability Hub project as part of a National Science Foundation-funded research initiative. Co-authored and co-presented the research abstract for the project.

DenAI Summit, Denver, CO

September, 2024

- Attended this AI-focused industry summit hosted in Denver, featuring talks by tech leaders, including CEOs and executives from Google and other major companies, start-ups, and the Denver mayor.

Skills

Languages & Frameworks:

Java, Python, HTML, CSS, JavaScript, SQL, R, ARM Assembly, Ruby, Flask, Flutter, and React

Artificial Intelligence & Data Science:

TensorFlow/Keras, SKLearnVectorStore, Llama3, Machine Learning, and Deep Learning

Soft Skills:

Communication, Problem-Solving, Leadership, Adaptability, Creativity, and Effective Time Management

Development Tools & Platforms:

Figma, Docker, Postman, Streamlit, Heroku, Swagger, MongoDB, Firebase Integration, VS Code, JetBrains IntelliJ, Eclipse, Repl.it [↗](#), and Jupyter Notebook

Version Control & Project Management:

Git/GitHub, ZenHub, Agile Methodologies

Personal learning:

Online Udemy courses, LeetCode, CodeAcademy, Coursera, and Khan Academy

Academic & Professional Memberships

- **Vice President** of UPE Honors Society (Upsilon Pi Epsilon)
- Girls Who Code (GWC)
- Association for Computing Machinery (ACM)
- Advanced Manufacturing Club (AMC)
- Colorado Uplift
- Denver Kids
- A+ Angels

Personal Interests

- Hackathons
- Building Computers
- Anime
- Weightlifting
- Muay Thai
- Animals
- Gymnastics

Relevant Courses Taken

Computer Science 1 (CS 1050) | **Computer Science 2** (CS 2050) | **Discrete Structures** (CS 2240) | **Computer Science Principles** (CS 1030) | **Computer Organization** (CS 1400) | **Computer Organization 2** (CS 2400) | **Principles of Programming Languages** (CS 3210) | **Algorithms & Algorithm Analysis** (CS 4050) | **Intro to Theory of Computation** (CS 3240) | **Operating Systems** (CS 3600) | **Calculus 1** (MTH 1410) | **Software Development Methods and Tools** (CS 3250) | **Probability and Statistics** (MTH 3210) | **Principles of Database Systems** (CS 3810) | **Software Engineering Practices** (CS 4360) | **Computer Security Offense and Defense** (CS 3755) | **Applied Methods in Linear Algebra** (MTH 3130) | **Web Application Development** (CS 3710) | **Computers, Ethics & Society** (PHI 3370) | **Deep Learning for Generating Educational Content** (CS 398F) | etc...

References

Multiple academic and professional references are available upon request.